

**Sandhya M 250701639-2025-CSE** ▾**S2****Started on** Friday, 11 September 2026, 9:28 AM**State** Finished**Completed on** Friday, 11 September 2026, 9:30 AM**Time taken** 1 min 45 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int n, i, j;
6      int a[100];
7
8      scanf("%d", &n);
9
10     for(i = 0; i < n; i++)
11         scanf("%d", &a[i]);
12
13     for(i = 0; i < n; i++)
14     {
15         for(j = i + 1; j < n; j++)
16         {
17             if(a[i] == a[j])
18             {
19                 printf("%d", a[i]);
20                 return 0;
21             }
22         }
23     }
24
25     return 0;
26 }
27

```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)

**Sandhya M 250701639-2025-CSE** ▾**S2****Started on** Friday, 11 September 2026, 9:30 AM**State** Finished**Completed on** Friday, 11 September 2026, 9:32 AM**Time taken** 1 min 34 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int n, i, j;
6      int a[100];
7
8      scanf("%d", &n);
9
10     for(i = 0; i < n; i++)
11         scanf("%d", &a[i]);
12
13     for(i = 0; i < n; i++)
14     {
15         for(j = i + 1; j < n; j++)
16         {
17             if(a[i] == a[j])
18             {
19                 printf("%d", a[i]);
20                 return 0;
21             }
22         }
23     }
24
25     return 0;
26 }
27

```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



Sandhya M 250701639-2025-CSE ▾

S2**Started on** Friday, 11 September 2026, 9:32 AM**State** Finished**Completed on** Friday, 11 September 2026, 9:35 AM**Time taken** 3 mins 1 sec**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int T, n1, n2, i, j;
6
7      scanf("%d", &T);
8
9      while(T--)
10     {
11         scanf("%d", &n1);
12
13         int a[n1];
14
15         for(i = 0; i < n1; i++)
16             scanf("%d", &a[i]);
17
18         scanf("%d", &n2);
19
20         int b[n2];
21
22         for(i = 0; i < n2; i++)

```



```
22 // scanf("%d", &a[i]);
23     scanf("%d", &b[j]);
24
25     i = 0;
26     j = 0;
27
28     while(i < n1 && j < n2)
29     {
30         if(a[i] == b[j])
31         {
32             printf("%d ", a[i]);
33             i++;
34             j++;
35         }
36         else if(a[i] < b[j])
37         {
38             i++;
39         }
40         else
41         {
42             j++;
43         }
44     }
45
46     printf("\n");
47 }
48
49 return 0;
50 }
51
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.



[Back to Course](#)

**Sandhya M 250701639-2025-CSE** ▾**S2****Started on** Friday, 11 September 2026, 9:36 AM**State** Finished**Completed on** Friday, 11 September 2026, 9:39 AM**Time taken** 2 mins 38 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int T, n1, n2, i, j;
6
7      scanf("%d", &T);
8
9      while(T--)
10     {
11         scanf("%d", &n1);
12
13         int a[n1];
14
15         for(i = 0; i < n1; i++)
16             scanf("%d", &a[i]);
17
18         scanf("%d", &n2);
19
20         int b[n2];
21
22         for(i = 0; i < n2; i++)

```

```
22 // scanf("%d", &a[i]);
23 scanf("%d", &b[j]);
24
25 i = 0;
26 j = 0;
27
28 while(i < n1 && j < n2)
29 {
30     if(a[i] == b[j])
31     {
32         printf("%d ", a[i]);
33         i++;
34         j++;
35     }
36     else if(a[i] < b[j])
37     {
38         i++;
39     }
40     else
41     {
42         j++;
43     }
44 }
45
46 printf("\n");
47 }
48
49 return 0;
50 }
51
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.



[Back to Course](#)



Sandhya M 250701639-2025-CSE ▾

S2**Started on** Tuesday, 22 September 2026, 9:37 PM**State** Finished**Completed on** Tuesday, 22 September 2026, 9:41 PM**Time taken** 4 mins 14 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n, k, i, j;
5
6      scanf("%d", &n);
7
8      int A[n];
9
10     for (i = 0; i < n; i++)
11         scanf("%d", &A[i]);
12
13     scanf("%d", &k);
14
15     for (i = 0; i < n; i++) {
16         for (j = i + 1; j < n; j++) {
17             if (A[j] - A[i] == k) {
18                 printf("1");
19                 return 0;
20             }
21         }
22     }
23
24     printf("0");
25
26     return 0;
27 }
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓

	Input	Expected	Got	
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)

**Sandhya M 250701639-2025-CSE** ▾**S2****Started on** Tuesday, 22 September 2026, 9:43 PM**State** Finished**Completed on** Tuesday, 22 September 2026, 9:49 PM**Time taken** 5 mins 59 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n, k, i, j;
5      scanf("%d", &n);
6
7      int a[n];
8
9      for(i = 0; i < n; i++)
10         scanf("%d", &a[i]);
11
12     scanf("%d", &k);
13
14     for(i = 0; i < n; i++)
15         for(j = i + 1; j < n; j++)
16             if(a[j] - a[i] == k) {
17                 printf("1");
18                 return 0;
19             }
20
21     printf("0");
22     return 0;
23 }
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓

	Input	Expected	Got	
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)